Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2013–2026)

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Contents

Abstract

Background

Epilepsy affects approximately 60 million people worldwide. One third of patients remain drug-resistant, and 69% of deaths from generalized tonic-clonic seizures could be prevented with timely detection (Sveinsson et al., 2020). EEG-based prediction is effective but unsuitable for ambulatory use due to operational complexity. Wearable devices using autonomic and motion signals offer an alternative, but translating deep learning approaches from controlled settings to real-world deployment remains challenging.

Methods

Following the PRISMA guidelines and the Webster & Watson concept matrix framework, this systematic review synthesizes 13 primary studies published between 2013 and 2026. Nine studies address detection and four address forecasting. The studies were mainly evaluated based on algorithmic architecture, biosignal modality, validation methodology, sensitivity, and false alarm rate.

Results

Convulsive seizure detection achieves high sensitivity (71.6-100%) primarily through accelerometry. Only two systems achieve a FPR deemed acceptable in ILAE Phase 3 of 0.008-0.028/h: Spahr et al. (2025) (96% sensitivity, 0.0054/h) and Fine (2025) (100% sensitivity, 0.023/h). Although these studies meet the performance standards of Phase 3, they are not truly Phase 3 because they were only validated in EMU. ECG-based detection of focal seizures fails clinically due to excessive false alarm rates (Reintjes et al., 2025).

Seizure forecasting currently faces a performance ceiling. Across methods and modalities, AUC consistently plateaus around 0.74-0.75 (Meisel et al., 2020; Vieluf et al., 2025). Under leave-one-subject-out validation, only 43% of patients achieve better-than-chance performance. The patient-specific models achieve 100% success but require weeks to months of individual data, creating a cold-start problem for newly diagnosed patients.

Multi-modal approaches dominate research (69% of studies) yet the best detection result uses single-modality accelerometry. This suggests biological specificity matters more than sensor count. Wrist-worn devices predominate (62% of studies). The validation rigor is generally insufficient. Only 3 of 13 studies employ prospective designs, and 11 of 13 validate in hospital settings that likely underestimate real-world false alarm rates.

Conclusions

Wearable seizure detection using DL architectures is close to reaching clinical viability for nocturnal convulsive seizures, but translation to home settings is missing. Forecasting remains immature without established benchmarks or regulatory pathways. Three gaps impede progress regarding forecasting: lack of large-scale home validation, absence of standardized forecasting benchmarks, and no viable solutions yet for approximately 81% of patients with focal onset seizures (Ayub et al., 2020). In most studies false alarm rates remain the primary barrier for daytime detection. Future research should prioritize prospective validation in the environment they will be deployed in, address the cold-start

problem through adaptive personalization and identify novel biomarkers for focal seizures without motor manifestations.

1 Introduction

1.1 Problem Statement and Motivation

Epilepsy is a chronic neurological disorder affecting 60 million people worldwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveinsson et al. (2020) found that 69% of deaths due to a generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.

While reliable prediction is already possible through electroencephalography (EEG), its operational complexity renders it unsuitable for ambulatory use. To address the need for ambulatory detection, researchers are experimenting with utilizing autonomic system and motion signals to detect and predict seizures.

Translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the complex, non-linear temporal dynamics of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction addressing these exact limitations.

This review examines and assesses recent advances in time-series deep learning - particularly convolutional neural networks (CNNs) and long short term memory (LSTMs) - applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of which architectural choices, training paradigms (e.g., transfer learning, personalization), and evaluation frameworks meet the strict clinical requirements of high sensitivity and minimal false alarm rates (FAR) or false positive rates (FPR) in real-world deployment.

1.2 Key Research Question

How do different deep learning architectures and biosignal modalities, excluding EEG, compare in their ability to achieve an optimal trade-off between sensitivity and false alarm rate for home seizure monitoring?

1.3 Clinical Evaluation Frameworks

1.3.1 ILAE Phased Validation Framework

The clinical validity of seizure detection algorithms is assessed through the International League Against Epilepsy (ILAE) phased framework, which classifies studies into five validation phases (Beniczky & Ryvlin, 2018; Beniczky et al., 2021).

Phase 0 represents proof of concept. Phase 1 involves retrospective analysis with small samples.

Phase 2 requires minimum 10 patients with seizures and 15 recorded seizures.

Phase 3 denotes prospective, multi-center validation with at least 30 seizures from at least 20 patients achieving at least 90% sensitivity with acceptable FPR using real-time

detection recorded during daily activities, so Phase 3 excludes epilepsy monitoring unit (EMU) data.
Phase 4 represents real-world home validation.
Prospective trials (Phase 3) constitute the clinical gold standard, requiring locked algorithms, video-EEG reference standards, and testing on unseen data from multiple centers in real-world scenarios.

1.3.2 Study Design Classifications

This review distinguishes three study design types.
Retrospective designs often risk data leakage when random splits violate temporal order, allowing future data to inform training (Kalousios et al., 2024; Wong et al., 2023).
Pseudo-prospective validation is a subset of the Retrospective study design that enforces chronological order, training only on past seizures to test on subsequent ones, providing more realistic performance estimates. (Kalousios et al., 2024; Lu et al., 2025).
Prospective studies validate models by deploying locked algorithms in real-time on patients. This makes it the most unbiased testing framework.

1.3.3 Clinical Performance Benchmarks

The ILAE systematic review established performance benchmarks for generalized tonic-clonic seizures (GTCS, bilateral motor seizures affecting both sides of the brain with loss of consciousness) and focal-to-bilateral tonic-clonic seizures (FBTCS, seizures starting in one hemisphere and spreading to both). Validated devices achieve 90-96% sensitivity with false alarm rates of 0.2-0.67 per 24 hours (approximately 0.008-0.028/h) (Beniczky et al., 2021).
For ambulatory use at home, acceptable false alarm rates depend on user perspective. Patients and caregivers typically require 100% sensitivity and accept approximately one false alarm per week. Clinicians consider 90% sensitivity adequate and tolerate two false alarms per week to one per month (Sivathamboo et al., 2022).
The ILAE Phase 3 benchmark of 0.008-0.028/h represents the clinically validated performance standard for GTCS/FBTCS detection.
Currently, the ILAE recommends wearable devices only for GTCS/FBTCS detection in unsupervised patients where alarms can alert caretakers for intervention. This represents a weak recommendation based on moderate evidence. For all other seizure types, the ILAE does not recommend clinical use of currently available wearable devices because of their lacking performance (Beniczky et al., 2021).

2 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles (Tugwell & Tovey, 2021)). The methodological approach uses a multi-stage search and transparent extraction and synthesis procedures as outlined below.

1. **Search strategy (Multi-stage, 2015-2025):**
    - **Scoping:** Google Scholar for broad coverage and identifying candidate papers.

- **Formal queries:** IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
- **Citation chasing:** Backward and forward citation tracking on key papers to ensure completeness.

2. **Inclusion / Exclusion criteria:**

   - **Inclusion:** Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
   - **Exclusion:** EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram is provided in Figure 1.

3 Search Strategy

3.1 PICO-Based Search Methodology

The literature search was structured using the PICO framework (Patient/Problem, Intervention, Comparison, Outcome) to identify relevant studies on deep learning for non-EEG epilepsy detection. The following key terms were used in combination with Boolean operators (AND between P,I,C,O, OR in the categories themselves and NOT):

1. **P (Patient/Problem):** Epileptic seizures in need of detection or prediction.

   - **Keywords:** 'epilepsy', 'seizure', 'ictal', 'preictal'
   - **MeSH Terms (PubMed):** Epilepsy, Seizures

2. **I (Intervention):** The diagnostic method based on wearable biosignals and deep learning.

   - **Biosignal Keywords:** 'electrocardiogram', 'ECG', 'heart rate variability', 'HRV', 'photoplethysmography', 'PPG', 'electrodermal activity', 'EDA', 'accelerometer', 'ACC'
   - **Technology Keywords:** 'wearable', 'wearable device'
   - **Method Keywords:** 'deep learning', 'machine learning', 'convolutional neural network', 'CNN', 'long short-term memory', 'LSTM', 'neural network'
   - **MeSH Terms (PubMed):** Wearable Electronic Devices, Electrocardiography, Deep Learning

3. **C (Comparison):** The explicit exclusion of studies focused solely on electroencephalography (EEG) as the primary modality.

   - **Exclusion Keywords:** 'EEG', 'electroencephalography', 'iEEG', 'intracranial'
   - **Search Logic:** The `NOT` operator was applied to this group in all database searches.

4. **O (Outcome):** Metrics quantifying the performance of a detection or prediction system.

    - **Keywords:** ‘detection’, ‘prediction’, ‘sensitivity’, ‘specificity’, ‘false alarm rate’, ‘FAR’, ‘area under the curve’, ‘AUC’

3.2 Practical Filters and Reproducibility

Consistent with the overall review scope, searches were restricted to **2015-2025** and **journal articles** with few exceptions made regarding **conference proceedings**. All database searches were documented (query string, platform, and execution date). Results were deduplicated prior to screening.

3.3 Exploration

Due to the insufficient amount of papers gathered from systematic review an exploration was done on the already existing papers. The exploration consisted of graph tools to visualize backlinks from the 9 already found studies and gemini deep research feature to find relevant papers. Through this method an additional 4 papers were acquired bringing the total number of papers up to 13 and expanding the time frame to 2013-2026.

3.4 Prisma-style flow diagram

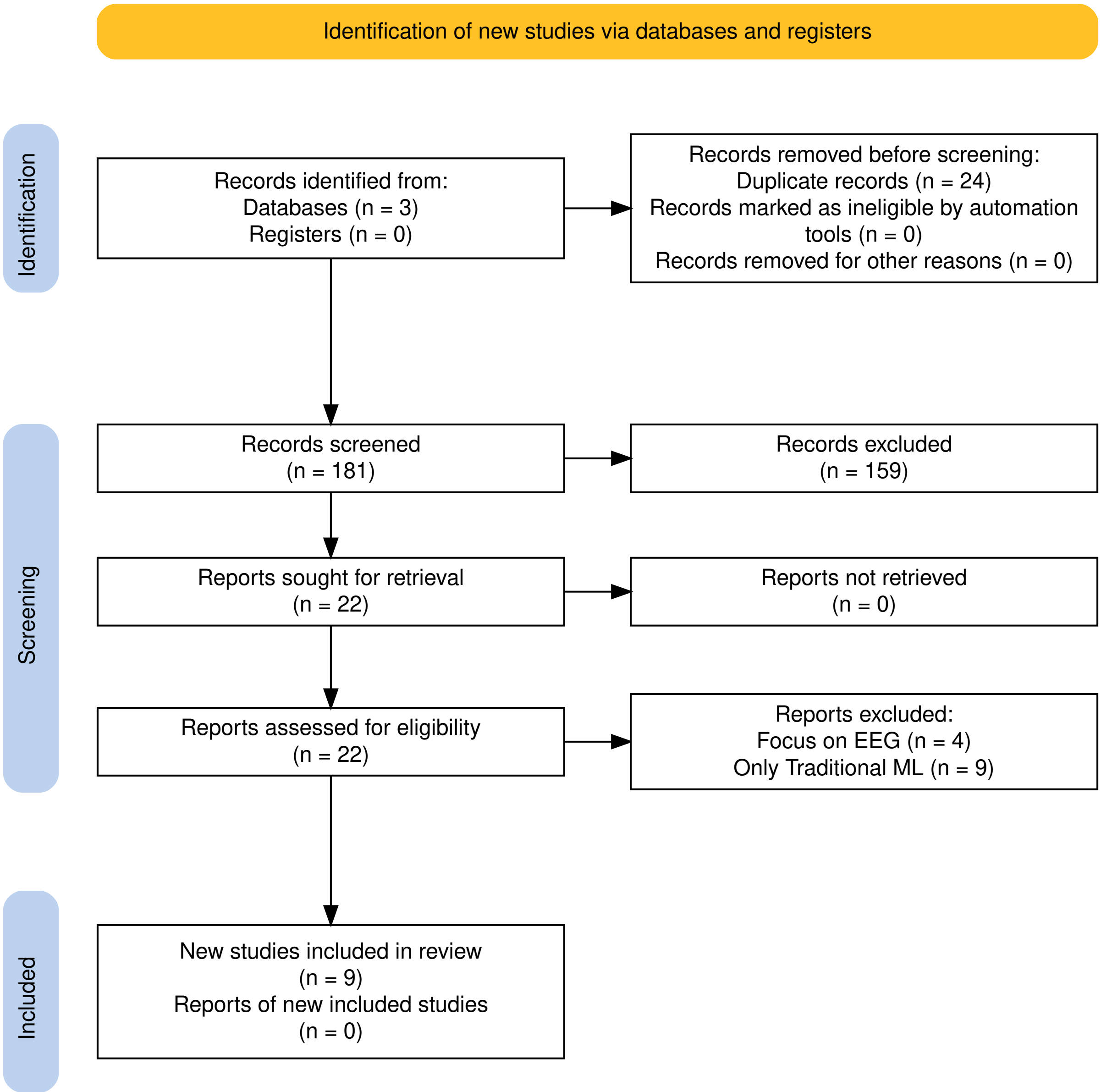

Figure 1: PRISMA flow diagram of the study selection process.

4 Tables

Table 1: Overview: Detection Studies

Study	Design	Sample	Device	Loc.	Para.	Mod.	Pers.	Algorithm	Sens	Spec	FPR
Spahr et al. 2025	Prospective	n=384, 49 CSs	Empatica E4 (ACC)	Wrist	Ens	1	Glb	Ensemble 1D CNN (30 models)	96%	–	0.0054/h
Reintjes et al. 2025	Retro	n=120, 856 szs	Single-lead ECG	Chest	Anom	1	Subj	Anomaly: Matrix Profile, MADRID, TimeVQVAE	2.44–82.79%	–	0.11–65.62/h
Fine 2025	Phase 1	n=15/3	6-axis band (ACC+Gyro)	Wrist	Feat	M	Glb	ANN (594 handcrafted feat)	100%Test, 96% complete	–	0.023/h
Dong et al. 2026	Prospective	n=68, 1846 szs	NightWatch	Arm	2-stg	M	Glb	CNN-LSTM + Attention	71.6%	–	0.165/h
Wang et al. 2025	–	n=28, 62 szs	Biovital-P1	Wrist	Feat	M	Glb	LSTM (40 hidden)	56.40% to 95.3%	–	0.364/h
Singh Rathore 2024	Case study	1 pt, 6 hrs	Wearable (EDA+ACC+HR)	–	Feat	M	CS	MLP (multi-modal features)	96.8%	94.8%^{prec}	–
Elemam et al. 2025	Cross-sect	Q:198, HRV:30	Camera PPG	Thumbs	Hyb	M	Glb	CNN + Rule-based fusion	95.1% Q, 93% HRV	97.06% Q	–
Borujeny 2013	3-pt	3 pts, 20 szs	MICAz ACC	Arm/thigh	Feat	1	Glb	ANN (time-domain feat)	85%	–	3 FA

Note: ACC = accelerometer, ANN = artificial neural network, CNN = convolutional neural network, CS = convulsive seizure / case study, ECG = electrocardiogram, EDA = electrodermal activity, HR = heart rate, HRV = HR variability, LSTM = long short-term memory, PPG = photoplethysmography, Sens = sensitivity, Spec = specificity, sz(s) = seizure(s), FPR = false positive rate, – = not reported. Para. = learning paradigm (Feat = feature-based, Ens = ensemble, Anom = anomaly detection, 2-stg = two-stage, Hyb = hybrid). Mod. = modality count (1 = single-modal, M = multi-modal). Pers. = personalization (Glb = global model, Subj = subject split, CS = case study). ^{prec} = precision (not specificity).

Table 2: Overview: Forecasting Studies

Study	Design	Sample	Device	Loc.	Para.	Mod.	Pers.	Algorithm	Sens	Spec	FPR
Vieluf et al. 2025	Retro	n=70, 5437 d	Embrace	Wrist	Hyb	M	Mix	DNN + harmonic features + diary	82%	67%	–
Meisel et al. 2020	LOSO CV	n=69, 452 szs	Empatica E4	Wrist/Ankle	E2E	M	Glb	LSTM (10 units)	51.2%	–	TiW 43.7%
Stirling 2021	Retro+pseudo	n=11, 136 szs	Fitbit	Wrist	Feat	M	Pat	LSTM+RF+LR ensemble (HRV feat)	–	–	AUC 0.74
Nasseri 2021	Retro	n=6	Empatica E4	Wrist	E2E	M	Tmp	LSTM 4-layer (128 hidden)	AUC 0.75 (SD 0.15)	–	TiW 0.9–7.2h/d
Ode et al. 2023	Retro	n=66, 85 szs	ECG	–	Anom	1	Pat	Self-Attentive AE	74%	–	0.85/h

Note: ACC = accelerometer, AE = autoencoder, CNN = convolutional neural network, DNN = deep neural network, ECG = electrocardiogram, EDA = electrodermal activity, LSTM = long short-term memory, LOSO = leave-one-subject-out, PPG = photoplethysmography, Sens = sensitivity, Spec = specificity, sz(s) = seizure(s), FPR = false positive rate, TiW = time in warning, AUC = area under curve, – = not reported. Para. = learning paradigm (Feat = feature-based, E2E = end-to-end, Anom = anomaly detection, Hyb = hybrid). Mod. = modality count (1 = single-modal, M = multi-modal). Pers. = personalization (Glb = global model, Pat = patient-specific, Mix = mixed, Tmp = temporal split).

Table 3: Architecture and Modality: Detection Studies

Study	Paradigm	Arch. Details	Modality Details	Fusion	Personalization	Trade-off Config	Deployment
Spahr et al. 2025	Ensemble	30 1D CNN models, 14 conv layers, quantile	ACC 32 Hz, Euclidean norm, 30 s	–	Global, tunable	86–100% @ 0.0054/h	Real-time (112 ms), on-device
Reintjes et al. 2025	Anomaly	Matrix-Profile, MADRID, TimeVQVAE-AD	ECG 256→8 Hz, bandpass	–	Subject split	2.44–82.79% @ 0.11–65.62/h	Offline, hospital
Fine 2025	Feature-based	ANN, 594 handcrafted features	ACC+Gyro 6-axis, 10 s intervals	Early	Global, hold-out	100% @ 0.023/h	Offline PC, EMU
Dong et al. 2026	Two-stage	Pre-screening + CNN-LSTM + Attention	ACM 11-12→20 Hz, PPG 100→20 Hz, 5 min	Early	Global, 10-fold CV	71.6% @ 0.165/h	Real-time, NightWatch, home
Wang et al. 2025	Feature-based	LSTM (40 hidden) + ReLU + FC	ACC/GYR 50 Hz, SEMG 200 Hz, EDA 4 Hz, 4 s	Early	Global, hold-out	56.4–95.3% @ 0.364/h	Real-time, hospital
Singh 2024	Feature-based	MLP, multi-modal features	EDA, ACC, HR/HRV, 25k points	Early	Single pt (CS)	96.8% @ 94.8% prec	Real-time, cloud?
Elemam 2025	CNN-based	CNN for HRV, CNN for Audio (separate)	PPG 250 Hz, Audio 10 s	No fusion	Global, unclear	95.1% @ 97.06% spec	Real-time, hospital
Borujeny 2013	Feature-based	ANN, time-domain features	ACC 2D 3 Hz, 9 s	–	Global, 3 pts	85% @ 3 FA	Real-time, server (316 mW)

Note: Paradigm: Feature-based = handcrafted features, Ensemble = multiple models, Two-stage = sequential processing, Anomaly = unsupervised detection, CNN-based = convolutional approach. Arch. Details: architecture-specific components. Fusion: – = single-modal, Early = feature-level, No fusion = parallel separate models. Personalization: Global = population model, Subject split = train/test separated by subject. Trade-off Config: operating points with sensitivity @ FAR. Deployment: real-time capability, processing location, setting. ACC = accelerometer, ECG = electrocardiogram, PPG = photoplethysmography, EDA = electrodermal activity, HR = heart rate, HRV = HR variability, SEMG = surface electromyography, EMU = epilepsy monitoring unit.

Table 4: Architecture and Modality: Forecasting Studies

Study	Paradigm	Arch. Details	Modality Details	Fusion	Personalization	Trade-off Config	Deployment
Vieluf et al. 2025	Hybrid	DNN + harmonic modeling (24-h)	EDA 4 Hz, ACC 32 Hz, Temp 1 Hz, 10 min	Early	Mixed, 5-fold + LOO	82% sens, 67% spec @ F1 0.81	Offline MATLAB, home
Meisel et al. 2020	End-to-end	LSTM only (10 units)	EDA/ACC/BVP/Temp →4 Hz, 30 s	Early	Global, LOSO	51.2% @ IoC 14.1%	Offline, hospital
Stirling 2021	Feature-based	LSTM+RF+LR ensemble, LR combines	HR 5 s, steps/sleep 1 min, hourly/daily	Decision	Patient-specific, weekly	AUC 0.74, 100% pts	Home (Fitbit), app
Nasseri 2021	End-to-end	LSTM 4-layer, 128 hidden + FFT channels	ACC/BVP/EDA/Temp/HR →128 Hz, 60 s	Early (17-ch)	Temporal split	AUC 0.75 (SD 0.15) @ TiW 0.9–7.2h/d	Offline, cloud, home
Ode et al. 2023	Anomaly	Self-Attentive AE (attention)	ECG (RRI only), 45 s	–	Patient-specific (99% CL)	74% @ 0.85/h (99% CL)	Real-time, cloud, hospital

Note: Paradigm: Feature-based = handcrafted features, End-to-end = learns from raw data, Hybrid = combines both, Anomaly = unsupervised detection. Arch. Details: architecture-specific components. Fusion: – = single-modal, Early = feature-level, Decision = classifier output. Personalization: Global = population model, Patient-specific = individualized, Temporal = time-split within patient, LOSO = leave-one-subject-out, LOO = leave-one-out, Mixed = combination of approaches. Trade-off Config: operating points with sensitivity @ FAR. Deployment: real-time capability, processing location, setting. ACC = accelerometer, ECG = electrocardiogram, EDA = electrodermal activity, BVP = blood volume pulse, HR = heart rate, TEMP = temperature, FFT = fast Fourier transform, CL = confidence limit, AUC = area under curve, TiW = time in warning, IoC = Improvement over Chance.

Table 5: Performance Metrics Reporting Across Studies

Metric	Detection (n=9)	Forecasting (n=4)	Reported By
Sensitivity/Recall	8/8	1/4	Spahr, Reintjes, Fine, Dong, Wang, Singh Rathore, Elemam, Borujeny, Ode[a]
Specificity	3/9	0/4	Elemam, Vieluf, Wang
FPR/FA rate	8/9	0/4	Spahr, Reintjes, Fine, Dong, Wang, Singh Rathore, Elemam, Borujeny, Nasseri, Ode
AUC-ROC	3/9	3/4	Dong, Reintjes, Ode, Nasseri, Stirling
Detection Latency	4/9	1/4	Spahr, Fine, Elemam, Borujeny, Nasseri
Patient Success Rate	2/9	3/4	Reintjes, Meisel, Stirling, Nasseri, Ode
Precision/PPV	4/9	1/4	Dong, Singh Rathore, Elemam, Ode
IoC / Time in Warning	0/9	3/4	Meisel, Stirling, Nasseri

5 Results

This systematic review analyzed 13 primary studies published between 2013 and 2026. The evidence base comprises 9 detection studies and 4 forecasting studies with a total of 912 participants. Table 1 and Table 2 present the study matrices for detection and forecasting approaches, respectively. Table 3 and Table 4 provide architectural details, and Table 5 summarizes the metrics reported across studies.

5.1 Study Characteristics

The 13 studies included in this review all vary in sample size and study design to a great degree. Detection studies enrolled a median of 44 participants (range 1 to 384), while forecasting studies enrolled a median of 11 participants (range 6 to 70).

Three studies employed prospective designs. Spahr et al. (2025) conducted a multi-center prospective study across eight EMUs. Dong et al. (2022) prospectively monitored patients at home for up to three months. Fine (2025) prospectively recorded data from 15 patients in an EMU for tonic seizure detection. The remaining studies used retrospective analysis of existing datasets or small-scale feasibility designs.

Research activity on DL models for detection has accelerated over time. Three forecasting studies were published between 2020 and 2022. Nine detection studies were published between 2023 and 2026.

Seven studies were conducted in hospital or EMU settings. Five studies included home or ambulatory validation and Elemam et al. (2025) used nighttime recordings captured at homes and in-hospital.

Four studies used a sample size of less than 20 participants. The smallest sample size was used by Singh Rathore et al. (2024). This is a single-patient case study. Also Borujeny et al. (2013) evaluated only 3 patients. Small sample sizes limit the reliability of performance estimates and generalizability.

5.2 Modality Performance

Detection and forecasting approaches use different biosignal modalities. Detection studies prioritize movement sensors, while forecasting studies emphasize autonomic nervous system indicators.

5.2.1 Modality Usage

Accelerometer data is the most prevalent modality appearing in nine of 13 studies (69%). Six detection studies and three forecasting studies use accelerometry (ACC). Electrodermal activity was captured in five studies (38%). Electrocardiogram (ECG) or heart rate variance (HRV) data appears in five studies. Photoplethysmography (PPG) appears in four studies. Gyroscope data appears in three studies, all in detection applications.
Multi-modal approaches dominate the evidence base. Nine of thirteen studies (69%) use multiple sensor types. Four studies use single-modality approaches: Spahr et al. (2025) (ACC only), Reintjes et al. (2025) (ECG only), Borujeny et al. (2013) (ACC only), and Ode et al. (2023) (ECG, HRV only).

5.2.2 Sensor Location

Wrist-worn devices appear in eight studies (62%). Arm placement appears in three studies. Chest placement appears in one study. Thigh placement appears in one study. Borujeny et al. (2013) used multiple sensor locations (right arm, left arm, left thigh).
Wrist placement was preferred because of patient acceptance and the best availability of consumer devices. The Empatica E4 was used in three studies. The Embrace watch and a Fitbit smartwatch both appear in one study each.

5.2.3 Detection Performance by Modality

Single-modality accelerometer approaches achieve strong performance. Spahr et al. (2025) reported 96% sensitivity with 0.0054/h FPR using only ACC data from the Empatica E4. This was the lowest FPR among all detection studies. Borujeny et al. (2013) reported 85% sensitivity using 2D ACC, though with only three patients.
ECG-based detection for focal seizures is currently clinically non-viable due to unacceptable false alarm rates. Reintjes et al. (2025) compared three anomaly detection methods on single-lead ECG. The sensitivity ranged from 2.44% to 82.79%, while FPR ranged from 0.11/h to 65.62/h depending on method and configuration. The lowest FPR of 0.11/h detected only 2.44% of seizures.
In addition GTCS/FBTCS detection using ECG and HRV achieved 74% sensitivity with 0.85/h FPR (Ode et al., 2023).
Wang et al. (2025) directly compared their modality performance. Attitude angle features (PITCH and ROLL) outperformed raw accelerometer and gyroscope data for seizure detection. The architecture achieved 56.4% to 95.3% sensitivity depending on the feature configuration. This finding suggests that feature engineering can enhance the value of standard movement sensors considerably.

5.2.4 Forecasting Performance by Modality

Forecasting studies consistently use autonomic signals. Nasseri et al. (2021) combined ACC, PPG, EDA, temperature, and HR to achieve AUC 0.75 (SD 0.15). Meisel et al. (2020) used EDA, PPG, temperature, and ACC from the Empatica E4 to achieve 51.2% sensitivity with IoC 14.1% in LOSO validation.
Vieluf et al. (2025) combined EDA, ACC, and temperature with diary data. This study demonstrated that 24-hour harmonic patterns in EDA and accelerometry contain predictive information above chance.

Multi-modal approaches are common but not universally superior. Single-modality ACC (Spahr et al. (2025)) achieves the best FPR in the included detection literature. Forecasting consistently requires multi-modal autonomic data, but AUC-ROC plateaus around 0.75 regardless of modality combination.

5.3 Architectures and Personalization

5.3.1 Learning Paradigms

Deep learning approaches differ widely across studies. Feature-based approaches that extract handcrafted features for classifiers appear in four studies. Fine (2025) extracted 594 features from 6-axis ACC and gyroscope data for an ANN classifier. The large feature set raises overfitting concerns given the small sample size (n=15), and no study directly compares feature-based to end-to-end approaches. Wang et al. (2025) engineered attitude angle features for an LSTM network with 40 hidden units and Singh Rathore et al. (2024) used multi-modal features for an MLP classifier in a single-patient case study.

End-to-end learning appears in two forecasting studies. Meisel et al. (2020) used an LSTM with 10 units operating directly on raw multi-modal wrist data and Nasseri et al. (2021) used a 4-layer LSTM with 128 hidden units per layer receiving fast fourier transform (FFT) channels of raw sensor data.

Two studies employ hybrid approaches. Vieluf et al. (2025) combined a DNN with handcrafted harmonic features extracted from 24-hour patterns, while Elemam et al. (2025) used separate CNNs for HRV and audio data without fusion between modalities.

Spahr et al. (2025) trained 30 separate 1D CNN models with quantile aggregation, achieving 96% sensitivity with 0.0054/h FPR. The model called Episave operates on 3D-ACC data at 32 Hz using 30-second windows. Dong et al. (2022) used a two-stage architecture with threshold-based pre-screening followed by a CNN-LSTM model.

Reintjes et al. (2025) and Ode et al. (2023) both used anomaly detection. Reintjes et al. (2025) compared three unsupervised anomaly detection methods (Matrix Profile, MADRID, TimeVQVAE) on single-lead ECG data. Ode et al. (2023) used a self-attentive autoencoder for patient-specific anomaly detection on ECG and HRV data.

5.3.2 Architecture Patterns by Application

Deep learning architectures differ substantially between detection and forecasting applications. Forecasting studies concentrate on LSTM-based approaches while detection research is more diverse but CNN-based approaches dominate.

Spahr et al. (2025) implemented an ensemble of 30 CNN models with 14 convolutional layers each. Elemam et al. (2025) used separate CNNs for HRV and audio processing. Traditional neural networks appear in earlier studies, with Fine (2025) using an ANN with 594 handcrafted features achieving 100% sensitivity.

In three out of the five forecasting studies LSTM architectures were used. Meisel et al. (2020) implemented a minimal LSTM with 10 hidden units operating directly on raw multi-modal wrist data while Nasseri et al. (2021) used a deeper 4-layer LSTM with 128 hidden units per layer. Stirling et al. (2021) combined LSTM with random forest and logistic regression in an ensemble. None of the forecasting studies use CNN-based approaches, probably because forecasting requires more focus on modeling temporal dependencies rather than spatial patterns.

Handcrafted features still remain competitive. Three of eight detection studies use feature-based approaches with strong performance. Fine (2025) achieved 100% sensitivity using 594 handcrafted features.
Window sizes vary substantially between applications. Detection studies use shorter windows ranging from 4 seconds to 5 minutes. Forecasting studies use longer windows ranging from 30 seconds to 60 minutes.

5.3.3 Personalization Strategies

Enabling personalization is a fundamental design choice. Global models train on population data and apply to all patients while patient-specific models train on individual patient data. Five studies use global model approaches, four studies use patient-specific approaches, and one study uses a mixed approach.

Global Models. Spahr et al. (2025) trained a global model on 384 patients with tunable parameters for FPR/sensitivity prioritization. Fine (2025) trained on 15 patients and tested on 3 patients and Dong et al. (2022) trained on 68 patients with 10-fold cross-validation. 198 questionnaire responses and 30 HRV recordings were used to train by Elemam et al. (2025). Meisel et al. (2020) trained a global model and tested with LOSO validation, achieving 62% patient success (43 of 69 patients above chance).

Patient-Specific Models. Nasseri et al. (2021) trained separate models for each of 6 patients achieving 83% patient success (5 of 6 patients), meanwhile achieving 100% patient success for hourly forecasting Stirling et al. (2021) doing weekly retraining of individual models on 11 patients. These rates are substantially higher than the 62% achieved by Meisel et al. (2020) using a global model with LOSO validation.
Anomaly detection models were trained by Ode et al. (2023) at the 99% confidence level for each of 66 patients. Singh Rathore et al. (2024) presented a single-patient case study also representing an individual model.
The biggest limitation in using patient-specific approaches is they require a mean of 14.6 months of data per patient in Stirling et al. (2021) or 6+ months per patient in Nasseri et al. (2021). This data requirement creates a cold start problem where new patients must wait weeks or months before the system becomes useful.

Mixed Approaches. The only study using a mixed approach is Vieluf et al. (2025). The study combined population-level 24-hour harmonic patterns with individual diary data. This hybrid achieved 82% sensitivity and 67% specificity.

5.3.4 Validation Approaches

Validation approaches vary across studies. Reintjes et al. (2025) used subject-split validation, training on one set of patients and testing on different patients to assess generalization.
Nasseri et al. (2021) used pseudo-prospective validation. Early data from each patient served as training while later data served as testing. This approach assesses temporal stability rather than generalization to new patients.
Meisel et al. (2020) used LOSO validation to assess global model performance on unseen patients, achieving 62% patient success.

5.3.5 Temporal Evolution

The earliest study, Borujeny et al. (2013), used traditional ANN approaches on MICAz accelerometer data in a 3-patient study. Studies published between 2020 and 2022 focused on LSTM-based forecasting architectures. Studies published between 2023 and 2026 employed more complex architectures including ensembles, hybrid methods, and attention mechanisms.
Note: There is a lot of variation regarding the mentioned architectural details. Some studies report complete architectural details while others provide minimal information. This variability complicates direct comparison and replication. Only two of eight detection studies provide individual patient results.

5.4 Performance Metrics

Detection and forecasting studies use different evaluation metrics reflecting their distinct clinical objectives. Detection studies emphasize sensitivity and FPR. Forecasting studies report AUC-ROC, improvement over chance (IoC), and time in warning (TiW).

5.4.1 Detection Performance

No study in this review has achieved true Phase 3 prospective validation in home settings The notable high-performing studies include Fine (2025) at 100% sensitivity, Spahr et al. (2025) at 96% sensitivity, Singh Rathore et al. (2024) at 96.8% sensitivity in a single-patient case study, and Elemam et al. (2025) at 95.1% sensitivity using HRV-based detection. Detection sensitivity across studies ranges from 38.0% to 100% (Table 1).
Only six of nine detection studies actually report FPR which is the most important metric beside sensitivity. The ILAE systematic review of Phase 3 studies reported FPR of 0.2-0.67 per 24 hours (approximately 0.008-0.028/h) for validated devices using home validation (Beniczky et al., 2021).
Two studies meet this performance range however both were conducted in an EMU setting. Spahr et al. (2025) conducted prospective multi-center validation on 384 patients and achieved 0.0054/h FPR and a sensitivity of 96%. The model uses an adjustable ensemble aggregation quantile parameter that allows trading sensitivity for false alarms. At one extreme, sensitivity can be raised to 100%. At the other extreme, the FPR can be reduced to 1/100 days at the cost of a lower sensitivity of 86%.
Fine (2025) achieved 0.023/h FPR and 100% sensitivity but is classified as Phase 1 because of the small independent test set. Only 10 seizures from 3 patients were used.
Three detection studies do not report FPR. Singh Rathore et al. (2024), Elemam et al. (2025), and Borujeny et al. (2013) lack standardized FPR reporting. This renders clinical assessment and comparison across studies impossible.

5.4.2 Forecasting Performance

Sensitivity in forecasting ranges from 51.2% to 82%. Individual study results are reported in Sections 5.2 and 5.3. Vieluf et al. (2025) achieved 82% sensitivity with 67% specificity, while Meisel et al. (2020) reported 51.2% sensitivity with LOSO validation.
Forecasting specific metrics are also included by Meisel et al. (2020) reporting a IoC of 14.1% and a TiW of 43.7%. Nasseri et al. (2021) reported TiW ranging from 0.9 to

7.2 h/day and Stirling et al. (2021) reported mean prediction time of 37 min for hourly forecasts and 3 days for daily forecasts.
Forecasting studies remain in earlier validation phases. Vieluf et al. (2025) used Phase IV clinical trial data but employed retrospective analysis. Meisel et al. (2020) and Nasseri et al. (2021) represent Phase 2-equivalent validation with moderate samples.
Forecasting AUC consistently falls between 0.74 and 0.80 regardless of architecture specifics across studies. This consistency suggests a performance ceiling for non-invasive forecasting approaches using wearable biosignals.

5.5 Detection, Forecasting, and Clinical Readiness

Traditionally detection and forecasting serve largely different clinical purposes.
Detection systems alert caregivers when a seizure occurs enabling intervention during or immediately after the seizure. It is most valuable for generalized tonic-clonic or focal to bilateral tonic-clonic seizures where caregiver response can prevent injury or SUDEP.
In contrast, forecasting systems provide advance warning of increased seizure likelihood. This allows patients to modify activities, take rescue medication, or avoid risky situations. Forecasting could improve quality of life by reducing the uncertainty of when seizures might occur.

5.5.1 Maturity Comparison

Two detection studies achieve FPR performance comparable to Phase 3 studies in the ILAE systematic review. Several commercial devices exist for detection including the Embrace watch with FDA 510(k) clearance and NightWatch with CE approval in Europe. Still no forecasting study meets a clear clinical benchmark. No device has regulatory approval for seizure forecasting.
While detection has a framework for classifying (ILAE 5 Phases by Beniczky and Ryvlin (2018)), forecasting lacks equivalent standardized guidelines. Forecasting AUC consistently falls between 0.74 and 0.80 across studies, suggesting a performance ceiling for non-invasive approaches, but the clinical acceptability of this performance level remains unclear.
Sample size disparities also clearly reflect the maturity difference. Detection studies have a median sample size of 66 participants compared to 40 for forecasting studies. The largest detection study enrolled 384 patients across eight centers, while the largest forecasting study enrolled 70 patients.

5.5.2 Commercial Device Status

Six of 13 studies use commercially available devices. The Empatica E4 research wristband appears in three studies. The Embrace watch appears in one study. The NightWatch armband appears in one study. Fitbit smartwatches appear in one study.
The Embrace watch has FDA 510(k) clearance for seizure detection. NightWatch has CE approval in Europe. However, no device has regulatory approval specifically for seizure forecasting. Forecasting devices would need to establish clinical utility and safety through dedicated trials.
Seven studies use research prototypes without clear commercialization pathways. These prototypes demonstrate technical feasibility but may require substantial engineering for commercial deployment.

5.5.3 Real-World Deployment

Six studies include home or ambulatory validation. Dong et al. (2022) achieved the most extensive home validation with 788 overnight recordings over three months. Stirling et al. (2021) achieved the longest monitoring duration with a mean of 14.6 months per patient. Nasseri et al. (2021) achieved the most rigorous ambulatory validation with concurrent invasive EEG confirmation via the RNS system. However, only six patients were studied. Hospital settings often fail to capture the diversity of real-world activities that cause false alarms. Exercise, cooking, household chores, and other activities present challenges not encountered in controlled hospital environments.

5.5.4 Deployment Architecture

Three studies report on-device or real-time deployment capability. Cloud-based processing also appears in three studies.

Spahr et al. (2025) reported 112 ms inference time suitable for real-time processing, but did not specify the device used. Borujeny et al. (2013) reported 316 mW power consumption for server-based processing. Dong et al. (2022) deployed on the NightWatch armband with real-time processing.

Nasseri et al. (2021) and Ode et al. (2023) both used cloud-based approaches.

Cloud processing may introduce latency and connectivity issues. Offline processing appears in two studies without clear deployment pathways.

6 Discussion

This discussion synthesizes findings from 13 studies on wearable seizure detection and forecasting. The evidence reveals different maturity levels for detection versus forecasting, distinct technical requirements, and persistent barriers to clinical translation. The discussion is organized into four parts. First, clinical implications regarding device readiness and coverage gaps are examined. Second, technical trade-offs in modality selection, architecture design, and personalization strategies are analyzed. Third, limitations in the evidence base that constrain confidence in the findings are identified. Fourth, research priorities to advance the field toward clinical translation are proposed.

6.1 Clinical Implications

From a clinical perspective, three key findings emerge regarding the scope of seizure coverage, the readiness of detection devices, and the current state of forecasting technology.

6.1.1 Seizure Type Coverage Gap

Current wearable approaches address only a subset of seizure types. Accelerometer-based systems have shown to detect motor manifestations reliably and with acceptable sensitivity, but non-motor seizures remain difficult to detect. The ECG-based approach by Reintjes et al. (2025) examined whether autonomic signals could detect non-motor seizures. However, the observed sensitivity-FPR trade-off prevents clinical interpretation. The lowest false alarm rate detected fewer than 3% of seizures, while configurations achieving acceptable sensitivity generated false alarm burdens orders of magnitude above clinically acceptable thresholds. This fundamental trade-off demonstrates that single-lead ECG anomaly detection cannot currently provide clinically reliable focal seizure detection. Forecasting approaches may theoretically address this gap by relying on autonomic modalities rather than movement sensors. However, the consistent AUC ceiling around 0.75 suggests fundamental limits in predicting seizures without EEG biomarkers. As Ayub et al. (2020) found, approximately 81% of seizures are focal onset and 19% are generalized tonic-clonic, meaning current accelerometer-based wearable approaches cannot detect the majority of seizures (Ayub et al., 2020). This coverage gap represents a substantial unmet need that may require invasive monitoring or alternative biomarkers.

6.1.2 Detection Readiness

The detection literature demonstrates that clinically acceptable FPR is achievable with accelerometer-based approaches (see Section 5). Two studies achieve FPR performance comparable to Phase 3 studies in the ILAE systematic review, which reported 0.2-0.67 false alarms per 24 hours (approximately 0.008-0.028/h) (Beniczky et al., 2021). This suggests that motor seizure detection has reached sufficient maturity for clinical translation. However, this achievement remains concentrated in controlled EMU settings rather than real-world home environments, placing both in Phase 2.

The EMU-to-home translation gap represents a critical barrier. The two studies achieving benchmark FPR were validated in hospital settings where patient activities are restricted and controlled. Home environments introduce motion patterns from daily living that do not occur in EMUs, including exercise, household chores, and other routine activities. The substantial FPR increase observed in home validation (0.165/h versus 0.0054/h in EMU settings (Dong et al., 2026; Spahr et al., 2025)) suggests that algorithm robustness to real-world activities must become a research priority.

6.1.3 Forecasting Status

The forecasting literature shows a consistent performance ceiling around AUC 0.74 to 0.80 across different methods, modalities, and patient populations (see Section 5). This convergence suggests a fundamental limit on pre-seizure predictability using non-invasive wearable sensors. Unlike detection, where clear clinical benchmarks exist, forecasting lacks standardized validation guidelines. Without established performance targets, the clinical utility of AUC 0.75 remains uncertain.

The Patient-specific approach in Stirling et al. (2021) achieves 100% success (defined as performing better than chance) but requires on average 14.6 months of individual data collection before the system becomes useful. This cold-start problem represents a substantial barrier to clinical adoption, as newly diagnosed patients cannot wait more than a year for protection.

No device has regulatory approval specifically for seizure forecasting. Forecasting devices would need to establish clinical utility and safety through dedicated trials. The lack of clear benchmarks complicates the pathway to regulatory approval. In addition, the current research remains sparse and unfocused with significantly different forecasting windows and methods.

6.2 Technical Insights and Trade-offs

These clinical observations reflect underlying technical trade-offs in modality selection, architecture design, and personalization strategy that shape what is achievable with current wearable approaches.

6.2.1 Modality Paradox

The evidence reveals a paradox regarding modality selection. Multi-modal approaches dominate the research field, with 69% of studies using multiple sensor types. However, the study achieving the lowest FPR uses single-modality ACC signals. This finding suggests that sensor fusion is not always necessary for reliable detection, and that biological specificity of motor manifestations may be more important than modality count.

The physiological requirements differ between detection and forecasting. Detection can succeed with movement sensors alone because motor manifestations during seizures produce clear signals in ACC recordings. Forecasting requires autonomic modalities because pre-seizure changes must be detected before clinical symptoms appear. All forecasting studies incorporate EDA, ECG, or temperature data to capture these autonomic precursors. The contrast suggests that modality selection should be guided by the temporal phase of seizure activity being targeted.

6.2.2 Architecture Selection

Detection and forecasting applications employ different architectural patterns. Detection research shows methodological diversity including CNNs, LSTMs, ANNs, ensemble methods, and anomaly detection. This diversity may reflect the broader exploration phase of detection research. Forecasting studies concentrate heavily on LSTM variants, with three of four studies using this architecture. This convergence reflects the different temporal demands, as forecasting requires modeling longer-term dependencies than detection.

The persistence of competitive handcrafted feature approaches is noteworthy. Domain knowledge remains valuable despite the end-to-end learning trend, with engineered features achieving excellent performance in some studies. This suggests that purely data-driven approaches may not fully grasp the known physiological signatures of seizures or that current end-to-end approaches can be optimized further.

Ensemble methods achieve excellent results but require greater computational resources. The trade-off between ensemble complexity and deployability becomes important for commercial translation, as multi-model ensembles may be impractical for low-power wearable devices. No ensemble study provides direct comparison of their full architecture against single-model baselines. While Dong et al. (2026) demonstrates multimodal fusion benefits, none report ablation on ensemble size, making it impossible to quantify whether multi-model approaches substantially outperform simpler architectures.

No evidence supports that architectural complexity improves performance. A feature-based ANN achieved 100% sensitivity on independent test data, while a CNN-LSTM-attention architecture achieved only 76% sensitivity despite using a more sophisticated model. The most effective approaches used domain knowledge through handcrafted features rather than model complexity.

6.2.3 Personalization Dilemma

The personalization strategy presents a fundamental dilemma for clinical deployment. Global models enable immediate deployment without individual patient data, but only 43% of patients achieve performance above chance in LOSO validation in Meisel et al. (2020). This means most patients receive no meaningful protection from population-based systems, rendering them useless for clinical deployment.
Patient-specific approaches achieve 100% success rates but require substantial individual data collection. The cold-start problem means newly diagnosed patients must wait weeks or months before the system becomes useful. Mixed approaches combining population-level patterns with individual data may offer a middle path, but evidence remains limited to single studies.

6.2.4 The False Positive Challenge

Multi-modal sensor fusion does not automatically solve the false positive problem. Studies combining multiple modalities still report FPR above clinical targets. The challenge may stem from the fundamental similarity between seizure-related physiological changes and noise introduced by other activities or stressors independent of seizures.
Forecasting faces a different performance ceiling. The consistency of AUC around 0.75 across different methods and modalities suggests fundamental limits on pre-seizure predictability using non-invasive sensors. This ceiling may reflect genuine biological variability in pre-seizure states that cannot be resolved with current wearable modalities and requires further research on the exact biological mechanisms during an epileptic seizure.

6.3 Limitations of the Evidence Base

The strength of these conclusions is limited by evidence quality across four dimensions: sample size constraints, validation setting bias, inconsistent metrics reporting, and insufficient monitoring durations.

6.3.1 Sample Size Constraints

The evidence base is constrained by small sample sizes in multiple studies. Five studies have sample sizes below 20 participants, including single-patient case studies and 3-patient evaluations. Small samples limit the reliability of performance estimates and reduce confidence that results will generalize to broader epilepsy populations especially since epilepsy is in itself drastically different between multiple patients. The forecasting literature is particularly affected, with studies limited to 6-11 patients due to invasive EEG confirmation requirements.

6.3.2 Validation Setting Bias

Eight of 13 studies were conducted in hospital or EMU settings, creating a systematic bias that may underestimate real-world FPR. Hospital environments lack the diversity of daily activities that generate false alarms in home settings as discussed in Section 6.1.2.

6.3.3 Reporting Inconsistency

Inconsistent metrics reporting hinders cross-study comparison. Forecasting studies routinely report patient-level success rates, enabling assessment of which patients benefit from the technology. Detection studies rarely provide this granularity, with only two of nine studies reporting individual patient outcomes. This reporting gap prevents assessment of whether detection algorithms work reliably for all patients or only a subset. High aggregate sensitivity may mask substantial variability across patients that remains invisible without patient-level reporting. This is probable since epileptic seizures vary widely between patients. The model has a strong likelihood to focus on the biggest subset of patients with easier to detect and more similar seizure patterns. This would result in poor performance for patients with irregular seizure patterns.

6.3.4 Monitoring Duration and Cold-Start Problems

Patient-specific approaches require weeks to months of individual data before becoming useful, creating a cold-start problem for newly diagnosed patients. This requirement conflicts with the clinical need for immediate protection after diagnosis. While global models enable immediate deployment, they achieve substantially lower patient success rates and possibly low generalizability. The trade-off between immediate deployment and individualized performance remains unresolved without evidence from hybrid approaches.

6.3.5 Unvalidated Architectural Components

Complex components are adopted without demonstrating value. Attention mechanisms were used in only two studies, with no direct comparison showing their isolated contribution. Similarly, ensemble and multi-stage architectures lack comparisons to simpler baselines. The evidence shows no correlation between architectural complexity and performance, suggesting that model sophistication may be prioritized over necessary validation.

6.4 Future Directions

Building on the mentioned limitations, nine research priorities emerge to advance wearable seizure monitoring toward clinical utility.

First, forecasting needs standardized clinical benchmarks. Detection has established clinical benchmarks from the ILAE systematic review of Phase 3 studies, which reported FPR of 0.2-0.67 per 24 hours (approximately 0.008-0.028/h) for convulsive seizure detection (Beniczky et al., 2021). Forecasting lacks an equivalent standard. AUC values around 0.74 to 0.75 appear consistently across forecasting studies, but the clinical meaning of this performance level remains unclear. Metrics such as IoC and TiW need standardization. The field should establish minimum acceptable performance levels for forecasting devices to enable clinical translation.

Second, large-scale home validation is needed. Only two detection studies include prospective home validation. Hospital and EMU settings may not capture the diversity of real-world activities that cause false alarms. Exercise, cooking, household chores, and other daily activities present challenges not encountered in controlled environments. The 0.165/h FPR reported by Dong et al. (2022) in home settings exceeds the clinical benchmark, however it does come close. Future studies should prioritize prospective validation in intended use environments with sufficient sample sizes and duration.

Third, novel modalities are needed for non-convulsive seizures. Current approaches detect only motor seizures with convulsive manifestations. Focal aware seizures without motor signs remain undetectable with current wearable approaches. Spahr et al. (2025) achieved excellent performance for convulsive seizures using accelerometer data alone, but this approach cannot detect seizures without movement. Multi-modal autonomic approaches show promise but have not addressed non-convulsive seizure types. Novel biosignals, advanced feature engineering or further research into the exact mechanics of seizures may be required to detect the subtle physiological changes associated with focal seizures.

Fourth, adaptive personalization strategies could address the trade-off between easy deployment and individual performance. Global models enable immediate deployment but may not work equally well for all patients. Meisel et al. (2020) achieved 43% patient success with LOSO validation, indicating substantial inter-patient variability. Patient-specific models achieve higher individual success but require weeks to months of individual data. Stirling et al. (2021) required a mean of 14.6 months per patient. Nasseri et al. (2021) required 6+ months per patient (median 220 days). Mixed approaches that combine population-level patterns with individual adaptation, such as the method used by Vieluf et al. (2025), may offer a balance. Research should address the cold-start problem by developing models that can initialize from population data and adapt efficiently to individual patients.

Fifth, real-time processing and edge deployment require more attention. Most studies in this review ran algorithms offline on PCs or servers rather than on wearable devices. Clinical utility demands real-time detection and forecasting on-device to enable timely interventions and to eliminate the risk posed by connection loss. This requires optimization for low power consumption and limited memory. Spahr et al. (2025) is an exception, achieving 112 ms inference time suitable for smartwatch integration. Their model (Episave) uses only 3D-ACC data, enabling integration with low-cost off-the-shelf smartwatches. This approach could expand accessibility to seizure monitoring for people with epilepsy in low- and middle-income countries where specialized devices are less available. Future work should report computational requirements and demonstrate feasible deployment on target hardware.

Sixth, model explainability and interpretability are needed for clinical adoption. Deep learning architectures such as CNNs, LSTMs, and autoencoders operate as black boxes with limited transparency. Clinicians and patients need to understand what features trigger detections or forecasts to trust these systems. Nasseri et al. (2021) performed ablation studies to examine feature contributions, and Reintjes et al. (2025) analyzed feature importance. However, systematic interpretability frameworks are missing. Future research should integrate explainable AI techniques and provide interpretable outputs for clinical decision-making.

Seventh, regulatory pathways and clinical integration need consideration. None of the reviewed studies address FDA or CE marking requirements for medical devices. Integration with clinical workflows, liability considerations for failed detections or forecasts, and ethical implications remain unexplored. Device approval pathways require rigorous validation data, standardized manufacturing processes, and post-market surveillance plans. Research should engage with regulatory frameworks early in development to facilitate clinical translation.

Eighth, algorithm robustness and generalizability require multi-site validation. Most studies used single-site or single-country data, limiting generalizability across populations and healthcare systems. Multi-continental validation with diverse demographic, geographic, and clinical characteristics is needed. Cross-device validation is also lacking. Studies typically evaluate algorithms on specific hardware such as Empatica E4, but performance may differ on consumer devices like Apple Watch or Fitbit. Standardized datasets and benchmarks would enable fair comparison across approaches.

Ninth, long-term algorithm stability remains unaddressed. Seizure patterns evolve due to medication changes, disease progression, and physiological changes over time. Stirling et al. (2021) retrained models weekly, acknowledging performance drift. However, no study reports systematic evaluation of model validity over periods longer than several months. Longitudinal studies assessing how detection and forecasting performance decays over time are needed. Adaptive algorithms that can detect and compensate for performance degradation would improve clinical utility.

7 Conclusion

This systematic review analyzed 13 studies on wearable seizure detection and forecasting using deep learning approaches. The evidence base comprises 912 participants across nine detection studies and four forecasting studies published between 2013 and 2026.

The predominance of accelerometer-based detection reveals a fundamental constraint in wearable seizure monitoring. Six of nine detection studies rely on ACC signals, and the two studies achieving performance comparable to ILAE Phase 3 both use ACC. It is important to note that they still fall in phase 2 because of their lacking home validation. Spahr et al. (2025) achieved 0.0054/h FPR with single-modality ACC. Fine (2025) achieved 0.023/h FPR with a 6-axis ACC and gyroscope band. This convergence on a single modality reflects the biological reality that convulsive seizures produce distinct motor signatures that deep learning can extract reliably. However, it also means current wearable approaches can serve only the subset of patients with motor manifestations. Approximately 81% of seizures are focal onset and 19% are generalized tonic-clonic (Ayub et al., 2020). While generalized tonic-clonic seizures are less common, they carry a 27-fold increased SUDEP risk compared to non-GTC seizures (Sveinsson et al., 2020), making their detection a clinical priority.

Forecasting studies face a different constraint. The consistent AUC plateau around 0.74 to 0.75 across studies using different methods, modalities, and patient populations suggests a fundamental limit on pre-seizure predictability using non-invasive sensors. Unlike detection, where the motor event during a seizure produces a clear signal, forecasting must rely on autonomic precursors that may be inherently less specific. The autonomic nervous system responds to stress, exercise, sleep transitions, and other daily events. These obstacles may establish a performance ceiling for wearable forecasting that cannot be breached without EEG biomarkers or novel sensing modalities.

No study in this review has achieved all requirements for widespread clinical adoption. Commercial detection devices exist with FDA or CE approval, but the deep learning approaches reviewed here lack large-scale prospective validation. Forecasting faces an additional barrier. No device has regulatory approval for forecasting, and the field lacks established clinical benchmarks equivalent to the ILAE Phase 3 FPR standard. The performance achieved to date, while scientifically promising, may not be sufficient for regulatory approval or clinical utility. This suggests that DL architectures are not yet ready for clinical deployment.

The clinical pathway forward requires addressing three major gaps.

First, home validation at scale is needed to address validation setting bias. Eight of 13 studies in hospital settings would show much higher FAR in less controlled, more noisy real-world settings.

Second, patient-level reporting is required to determine detection reliability across diverse seizure types. Current aggregation hides performance variability, with only two studies reporting individual outcomes that may reveal algorithms work only for subsets of patients.

Third the trade-off between immediate deployment and individualized performance needs to be resolved. Patient-specific approaches require months of data collection while global models achieve substantially lower success rates. Mixed personalization strategies may balance these competing demands.

8 References

Ayub, N., Chiang, S., Moss, R., & Goldenholz, D. (2020). Natural history of generalized motor seizures: A retrospective analysis. *Seizure*, *80*, 109–112. DOI: 10.1016/j.seizure.2020.05.019 (cit. on pp. 1, 19, 24).

Beniczky, S., & Ryvlin, P. (2018). Standards for testing and clinical validation of seizure detection devices. *Epilepsia*, *59 Suppl 1*, 9–13. DOI: 10.1111/epi.14049 (cit. on pp. 2, 17).

Beniczky, S., Wiebe, S., Jeppesen, J., Tatum, W. O., Brazdil, M., Wang, Y., Herman, S. T., & Ryvlin, P. (2021). Automated seizure detection using wearable devices: A clinical practice guideline of the International League Against Epilepsy and the International Federation of Clinical Neurophysiology. *Clinical Neurophysiology*, *132*(5), 1173–1184. DOI: 10.1016/j.clinph.2020.12.009 (cit. on pp. 2, 3, 16, 19, 22).

Borujeny, G. T., Yazdi, M., Keshavarz-Haddad, A., & Borujeny, A. R. (2013). Detection of Epileptic Seizure Using Wireless Sensor Networks. *Journal of Medical Signals and Sensors*, *3*(2), 63–68. Retrieved January 5, 2026, from URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC3788195/ (cit. on pp. 12, 13, 16, 18).

Dong, C., van Dijk, J. P., Aarts, R. M., Wang, Y., & Long, X. (2026). Detection of nocturnal epileptic seizures using a wearable armband: A deep learning approach combining accelerometry and photoplethysmography signals. *Computer Methods and Programs in Biomedicine*, *273*, 109087. DOI: 10.1016/j.cmpb.2025.109087 (cit. on pp. 19, 20).

Dong, C., Ye, T., Long, X., Aarts, R. M., van Dijk, J. P., Shang, C., Liao, X., Chen, W., Lai, W., Chen, L., & Wang, Y. (2022). A Two-Layer Ensemble Method for Detecting Epileptic Seizures Using a Self-Annotation Bracelet With Motor Sensors. *IEEE Transactions on Instrumentation and Measurement*, *71*, 1–13. DOI: 10.1109/TIM.2022.3173270 (cit. on pp. 12, 14, 15, 18, 22).

Elemam, M. E., Elgamal, A., Elmenshawi, I., & Abdelkader, H. E. (2025). Automated validated tool for epileptic seizure detection using deep learning. *Journal of Intelligent Systems and Internet of Things*, *15*(1), 74–90. DOI: 10.54216/JISIoT.150107 (cit. on pp. 12, 14–16).

Fine, A. L. (2025). Detection is Key: Automated Tonic Seizure Detection With a Wearable Device. *Epilepsy Currents*, *25*(1), 36–38. DOI: 10.1177/15357597241293298 (cit. on pp. 1, 12, 14–16, 24).

Hussein, A. F., Arunkumar, N., Gomes, C., Alzubaidi, A. K., Habash, Q. A., Santamaria-Granados, L., Mendoza-Moreno, J. F., & Ramirez-Gonzalez, G. (2018). Focal and

Non-Focal Epilepsy Localization: A Review. *IEEE Access*, *6*, 49306–49324. DOI: 10.1109/ACCESS.2018.2867078 (cit. on p. 2).

Kalousios, S., Müller, J., Yang, H., Eberlein, M., Uckermann, O., Schackert, G., Polanski, W. H., & Leonhardt, G. (2024). ECG-based epileptic seizure prediction: Challenges of current data-driven models. *Epilepsia Open*, *10*(1), 143–154. DOI: 10.1002/epi4.13073 (cit. on p. 3).

Lu, S., Liu, L., Li, J., Chambers, J., Cook, M. J., & Grayden, D. B. (2025). Leveraging Channel Coherence in Long-Term iEEG Data for Seizure Prediction. *IEEE Journal of Biomedical and Health Informatics*, *29*(8), 5541–5548. DOI: 10.1109/JBHI.2025.3556775 (cit. on p. 3).

Meisel, C., El Atrache, R., Jackson, M., Schubach, S., Ufongene, C., & Loddenkemper, T. (2020). Machine learning from wristband sensor data for wearable, noninvasive seizure forecasting. *Epilepsia*, *61*(12), 2653–2666. DOI: 10.1111/epi.16719 (cit. on pp. 1, 13–17, 21, 23).

Nasseri, M., Pal Attia, T., Joseph, B., Gregg, N. M., Nurse, E. S., Viana, P. F., Worrell, G., Dümpelmann, M., Richardson, M. P., Freestone, D. R., & Brinkmann, B. H. (2021). Ambulatory seizure forecasting with a wrist-worn device using long-short term memory deep learning. *Scientific Reports*, *11*(1), 21935. DOI: 10.1038/s41598-021-01449-2 (cit. on pp. 13–18, 23).

Ode, R., Fujiwara, K., Miyajima, M., Yamakawa, T., Kano, M., Jin, K., Nakasato, N., Sawai, Y., Hoshida, T., Iwasaki, M., Murata, Y., Watanabe, S., Watanabe, Y., Suzuki, Y., Inaji, M., Kunii, N., Oshino, S., Khoo, H. M., Kishima, H., & Maehara, T. (2023). Development of an epileptic seizure prediction algorithm using R–R intervals with self-attentive autoencoder. *Artificial Life and Robotics*, *28*(2), 403–409. DOI: 10.1007/s10015-022-00832-0 (cit. on pp. 13–15, 18).

Reintjes, C., Hagenbeck, J. F., Ballo, M., Rahlmeier, T., Wolf, S. M., & Schoder, D. (2025). ECG-Based Detection of Epileptic Seizures in Real-World Wearable Settings: Insights from the SeizeIT2 Dataset. *Sensors*, *25*(24), 7687. DOI: 10.3390/s25247687 (cit. on pp. 1, 13–15, 18, 23).

Singh Rathore, S. P., Magotra, S., K V, S., Sonu Kumar, S., Kaur, G., & Singh, S. (2024). Development of a Multimodal Machine Learning Model for Seizure Detection Using Wearable Devices. *2024 1st International Conference on Advances in Computing, Communication and Networking (ICAC2N)*, 1422–1426. DOI: 10.1109/ICAC2N63387.2024.10895217 (cit. on pp. 12, 14–16).

Sivathamboo, S., Nhu, D., Piccenna, L., Yang, A., Antonic-Baker, A., Vishwanath, S., Todaro, M., Yap, L. W., Kuhlmann, L., Cheng, W., O'Brien, T. J., Lannin, N. A., & Kwan, P. (2022). Preferences and User Experiences of Wearable Devices in Epilepsy: A Systematic Review and Mixed-Methods Synthesis. *Neurology*, *99*(13), e1380–e1392. DOI: 10.1212/WNL.0000000000200794 (cit. on p. 3).

Spahr, A., Bernini, A., Ducouret, P., Baumgartner, C., Koren, J. P., Imbach, L., Beniczky, S., Larsen, S. A., Rheims, S., Fabricius, M., Seeck, M., Steinhoff, B. J., Beuchat, I., Dan, J., Atienza, D. A., Bardyn, C.-E., & Ryvlin, P. (2025). Deep learning-based detection of generalized convulsive seizures using a wrist-worn accelerometer. *Epilepsia*, *66 Suppl 3*, 53–63. DOI: 10.1111/epi.18406 (cit. on pp. 1, 12–16, 18, 19, 23, 24).

Stirling, R. E., Grayden, D. B., D'Souza, W., Cook, M. J., Nurse, E., Freestone, D. R., Payne, D. E., Brinkmann, B. H., Pal Attia, T., Viana, P. F., Richardson, M. P., & Karoly, P. J. (2021). Forecasting Seizure Likelihood With Wearable Technology.

Frontiers in Neurology, *12*, 704060. DOI: 10.3389/fneur.2021.704060 (cit. on pp. 14, 15, 17–19, 23, 24).

Sveinsson, O., Andersson, T., Mattsson, P., Carlsson, S., & Tomson, T. (2020). Clinical risk factors in SUDEP. *Neurology*, *94*(4), e419–e429. DOI: 10.1212/WNL.0000000000008741 (cit. on pp. 1, 2, 24).

Tugwell, P., & Tovey, D. (2021). PRISMA 2020. *Journal of Clinical Epidemiology*, *134*, A5–A6. DOI: 10.1016/j.jclinepi.2021.04.008 (cit. on p. 3).

Vieluf, S., Tomioka, S., Zhang, B., Krishnan, V., Bosl, W. J., Grinnell, T., & Loddenkemper, T. (2025). Seizure monitoring by combined diary and wearable data: A multicenter, longitudinal, observational study. *Epilepsia*, *66*(11), 4259–4271. DOI: 10.1111/epi.18550 (cit. on pp. 1, 13–17, 23).

Wang, J., Hu, D., Lai, X., Jiang, T., Jiang, T., Gao, F., Vidal, P.-P., & Cao, J. (2025). Epileptic Seizure Detection Based on Attitude Angle Signal of Wearable Device. *IEEE Transactions on Instrumentation and Measurement*, *74*. DOI: 10.1109/TIM.2025.3529058 (cit. on pp. 13, 14).

Webster, J., & Watson, R. T. (2002). Analyzing the past to prepare for the future: Writing a literature review. *MIS quarterly*, xiii–xxiii. Retrieved December 3, 2025, from URL: https://www.jstor.org/stable/4132319 (cit. on p. 3).

Wong, S., Simmons, A., Rivera-Villicana, J., Barnett, S., Sivathamboo, S., Perucca, P., Ge, Z., Kwan, P., Kuhlmann, L., Vasa, R., Mouzakis, K., & O'Brien, T. J. (2023). EEG datasets for seizure detection and prediction— A review. *Epilepsia Open*, *8*(2), 252–267. DOI: 10.1002/epi4.12704 (cit. on p. 3).